RDBMS

UNIT I

SECTION – A (3 MARKS)

1. **Define Database.**

- A **database** is an organised collection of data, so that it can be easily accessed and managed. This includes can organise data into tables, rows, columns, and index it to make it easier to find relevant information.
- The **main purpose** of the database is to operate a large amount of information by storing, retrieving, and managing data.
- Modern databases are managed by the database management system (DBMS).

2. **Define DBMS.**

- Database management system is a software which is used to manage the database. For example: MySQL, Oracle, etc are a very popular commercial database which is used in different applications.
- DBMS provides an interface to perform various operations like database creation, storing data in it, updating data, creating a table in the database and a lot more.
- It provides protection and security to the database. In the case of multiple users, it also maintains data consistency.

3. **List the Applications of DBMS.**

- Railway Reservation System
- Library Management System
- Banking
- Universities and colleges
- Credit card transactions
- Social Media Sites
- Finance

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- Military
  Online Shopping

- Human Resource Management

- Manufacturing

- Airline Reservation system

4. **What is RDBMS?**

- **RDBMS** stands for Relational Database Management System.and for all modern database systems like MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft
  Access.

- RDBMS is a database management system DBMS that is based on the relational model as introduced by E. F. Codd.

- Data is represented in terms of tuples (rows) in RDBMS. A relational database is the most commonly used database. It contains several tables, and each table has its primary key.

5. **What is a Distributed Database?**

- A distributed database is a collection of multiple interconnected databases, which are spread physically across various locations that communicate via a computer network.

- Databases in the collection are logically interrelated with each other. Often they represent a single logical database.

- They do not have any multiprocessor configuration.

6. **What is a Centralised Database?**

- A centralized database is stored at a single location such as a mainframe computer.

- It is maintained and modified from that location only and usually accessed using an internet connection such as a LAN or WAN.

- The centralized database is used by organisations such as colleges, companies, banks etc.

7. **List the Advantages of DBMS.**

DBMS makes it possible for end users to create, read, update and delete data in database. Some of these advantages are given below −

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- Reducing Data Redundancy
  Sharing of Data
- Data Integrity
- Data Security
- Privacy
- Backup and Recovery
- Data Consistency

8. List the types of Database.

- A **database** is an organised collection of data, so that it can be easily accessed and managed.

<u>The types of Database :</u>

- Centralised Database
- Distributed Database
- Relational Database
- NoSQL Database
- Cloud Database
- Object-oriented Databases ● Hierarchical Database.

9. What is a Network Database?

- A network database is a type of database model wherein multiple member records or files can be linked to multiple owner files and vice versa.
- The model can be viewed as an upside-down tree where each member information is the branch linked to the owner, which is the bottom of the tree.

10.What is Cloud Database?

- A cloud database is a database service built and accessed through a cloud platform.

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- It serves many of the same functions as a traditional database with the added flexibility of cloud computing.

  Users install software on a cloud infrastructure to implement the database.

11. What is Hierarchical Database?

- A hierarchical database is a data model in which data is stored in the form of records and organised into a tree-like structure, or parent-child structure, in which one parent node can have many child nodes connected through links between them.
- hierarchical data models have a treelike structure as their main characteristic.

12.What is a Relational Database?

- A relational database is a collection of data items with pre-defined relationships between them.
- These items are organized as a set of tables with columns and rows. Tables are used to hold information about the objects to be represented in the database.
- Each column in a table holds a certain kind of data and a field stores the actual value of an attribute.

13.List the advantages of a centralized Database.

- the data is in one place, there can be stronger security measures around it. So, the centralized database is much more secure.
- Data is easily portable because it is stored at the same place.
- The centralized database is cheaper than other types of databases as it requires less power and maintenance..

14.List the advantages of Distributed Database.

- The database can be stored according to the departmental information in an organisation. In that case, it is easier for an organisational hierarchical access.
- there were a natural catastrophe such as fire or an earthquake all the data would not be destroyed it is stored at different locations.
- It is cheaper to create a network of systems containing a part of the database. This database can also be easily increased or decreased.

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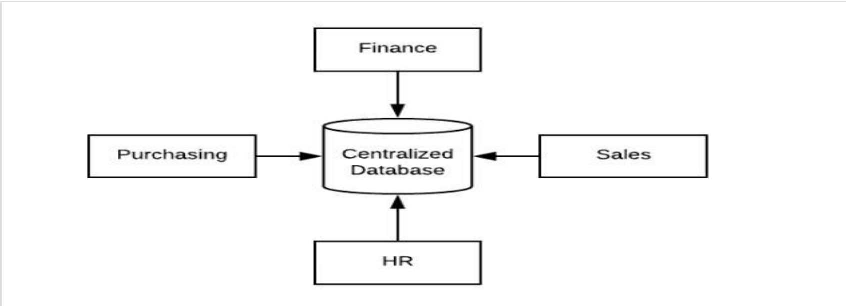
15.What is a Data Model?

- Data models define how the logical structure of a database is modeled. Data Models are fundamental entities to introduce abstraction in a DBMS.
  Data models define how data is connected to each other and how they are processed and stored inside the system.
- The very first data model could be flat data-models, where all the data used are to be kept in the same plane.
- Earlier data models were not so scientific, hence they were prone to introduce lots of duplication and update anomalies.

<u>**PART - B**</u>

1. Write a note on the Centralized database.

- A centralized database is stored at a single location such as a mainframe computer.
- It is maintained and modified from that location only and usually accessed using an internet connection such as a LAN or WAN.
- The centralized database is used by organizations such as colleges, companies, banks etc.



Advantages

- The data integrity is maximised as the whole database is stored at a single physical location. This means that it is easier to coordinate the data and it is as accurate and consistent as possible.
- The data redundancy is minimal in the centralized database.
- Since all the data is in one place, there can be stronger security measures around it. So, the centralized database is much more secure.
- Data is easily portable because it is stored at the same place.

Disadvantages

- Since all the data is at one location, it takes more time to search and access it. If the network is slow, this process takes even more time.
- There is a lot of data access traffic for the centralized database. This may create a bottleneck situation.
- Since all the data is at the same location, if multiple users try to access it simultaneously it creates a problem. This may reduce the efficiency of the system.
- If there are no database recovery measures in place and a system failure occurs, then all the data in the database will be destroyed

2. Write a note on the Distributed database.

- A distributed database is a collection of multiple interconnected databases, which are spread physically across various locations that communicate via a computer network.

Features

- Databases in the collection are logically interrelated with each other. Often they represent a single logical database.
- Data is physically stored across multiple sites. Data in each site can be managed by a DBMS independent of the other sites.
- The processors in the sites are connected via a network. They do not have any multiprocessor configuration.
- A distributed database is not a loosely connected file system.
- A distributed database incorporates transaction processing, but it is not synonymous with a transaction processing system.

Advantages

- **Modular Development** − If the system needs to be expanded to new locations or new units, in centralized database systems, the action requires substantial efforts and disruption in the existing functioning.
- **More Reliable** − In case of database failures, the total system of centralized databases comes to a halt.
- **Better Response** − If data is distributed in an efficient manner, then user requests can be met from local data itself, thus providing faster response.
- **Lower Communication Cost** − In distributed database systems, if data is located locally where it is mostly used, then the communication costs for data manipulation can be minimized. This is not feasible in centralized systems.

3. Explain Significance of DBMS.

- **Reduces data redundancy:** Data redundancy occurs when end-users use the same data in different locations. Using a DBMS, a user can store data in a centralized place, which reduces the requirement of saving the same data in many locations.
- **Ensures data security:** A DBMS ensures that only authorized people have access to specific data. Instead of giving all users access to all the data, a DBMS allows you to define who can access what.
- **Eliminates data inconsistency:** As data gets stored in a single repository, changing one application does not affect the other applications using the same set of details.
- **Ensures data sharing:** Using a database management system, users can securely share data with multiple users. As DBMS has a locking technology, it prevents data from being shared by two people using the same application at the same time.
- **Maintains data integrity:** A DBMS can have multiple databases, making data integrity essential for digital businesses. When a database has consistent information across databases, end-users can leverage its advantages.

- **Ensures data recovery:** Every DBMS ensures backup and recovery and end-users do not manually backup data. Having a consistent data backup helps to recover data quickly.
- **Low maintenance cost:** The initial expense for setting up a DBMS is high, but its maintenance cost is low.
- **Saves time:** Using a DBMS, a software developer can develop applications much faster.
- **Allows multiple user interfaces:** A DBMS allows different user interfaces as application program interface and graphical user interface.

4. Write a note on Applications of DBMS.

- **Railway Reservation System** − The railway reservation system database plays a very important role by keeping record of ticket booking,
- **Library Management System** − Now-a-days it's become easy in the Library to track each book and maintain it because of the database. DBMS used to maintain all the information related to book issue dates, name of the book, author and availability of the book.
- **Banking** − Banking is one of the main applications of databases. There will be a thousand transactions through banks daily This is all possible just because of DBMS that manages all the bank transactions.
- **Universities and colleges** − Now-a-days examinations are done online. So, the universities and colleges are maintaining DBMS to store Student's registrations details, results, courses and grade all the information in the database.
- **Credit card transactions** − The purchase of items and transactions of credit cards are made possible only by DBMS. information that all are secured through DBMS.
- **Social Media Sites** − All the information related to the users are stored and maintained with the help of DBMS.
- **Finance** − There are lots of things to do with finance like storing sales, holding information and finance statement management etc. these all can be done with database systems.
- **Military** − In military areas the DBMS is playing a vital role. Military keeps records of soldiers and it has so many files that should be kept secure and safe. DBMS provides a high security to military information.
- **Online Shopping** − Now-a-days we all do Online shopping without wasting the time by going shopping with the help of DBMS. The products are added and sold only with the help of DBMS like Purchase information, invoice bills and payment.
- **Human Resource Management** − The management keeps records of each employee's salary, tax and work through DBMS.

5. Differentiate between DBMS and RDBMS.

DBMS	RDBMS

DBMS applications store **data as file**.	RDBMS applications store **data in a tabular form**.
In DBMS, data is generally stored in either a hierarchical form or a navigational form.	In RDBMS, the tables have an identifier called primary key and the data values are stored in the form of tables.
Normalization is not present in DBMS.	**Normalization is** present in RDBMS.
DBMS does **not apply any security** with regards to data manipulation.	RDBMS **defines the integrity constraint** for the purpose of ACID
DBMS uses a file system to store data, so there will be **no relation between the tables**.	Data values are stored in the form of tables, so a **relationship** between these data values will be stored in the form of a table as well.
DBMS has to provide some uniform methods to access the stored information.	The RDBMS system supports a tabular structure of the data and a relationship between them to access the stored information.
DBMS **does not support distributed databases**.	RDBMS **supports distributed databases**.

DBMS is meant for small organizations and **deal with small data**. It supports **single user**.	RDBMS is designed to **handle large amounts of data**. it supports **multiple users**.

6. **Write a note on Relational and Object Oriented Database.**

<u>Relational Database</u>

- A relational database is a collection of information that organizes data points with defined relationships for easy access.
- In the relational database model, the data structures -- including data tables, indexes and views -- remain separate from the physical storage structures, enabling database administrators to edit the physical data storage without affecting the logical data structure.
- relational databases are used to organize data and identify relationships between key data points.
- They make it easy to sort and find information, which helps organizations make business decisions more efficiently and minimize costs.
- They work well with structured data.
- The data tables used in a relational database store information about related objects.
- Each row holds a record with a unique identifier -- known as a key -- and each column contains the attributes of the data.
- Each record assigns a value to each feature, making relationships between data points easy to identify.

<u>OOD</u>

- An object-oriented database is a collection of object-oriented programming and relational databases.

- There are various items which are created using object-oriented programming languages like C++, Java which can be stored in relational databases, but object-oriented databases are well-suited for those items.
- An object-oriented database is organized around objects rather than actions, and data rather than logic.

Complex data and a wider variety of data types compared to MySQL data types.

- Easy to save and retrieve data quickly.
- Seamless integration with object-oriented programming languages.
- Easier to model the advanced real world problems.
- Extensible with custom data types.

7. Write a note on characteristics and Advantages of DBMS.

<u>Characteristics</u>

1. DBMS for storing and manipulating the information uses a digital repository that is established on a server.
2. There are ACID properties within this system for maintaining data in a healthy state in case of any failures.
3. A clear and logical view of systems used for data manipulation is provided by DBMS.
4. DBMS facilitates in minimizing the complex relationship between data.
5. This system is composed of automatic backup and recovery procedures.
6. Data can be viewed from different aspects in accordance with user requirements via DBMS.
7. DBMS provides a complete security of data.

<u>Advantages</u>

Avoids Database Redundancy

- Data redundancy involves maintaining several copies of the same data.
- It records all data in one single file of the database which saves our storage space and enhances update as well as retrieval speed.

Data Abstraction

- Data abstraction is one of the key characteristics of DBMS.
- It involves hiding the complexities of data from basic users.

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Improved Data Sharing

- DBMS has improved the sharing of data by end-users via developing a friendly environment.
- Data can be easily shared among multiple users within the organization by authorized users. .

Enhanced Security Of Data

- Database management systems provide such a framework where there is a better enforcement of data privacy and security policies.

Minimize Data Inconsistency

- DBMS avoids the inconsistency of data by controlling the data redundancy.
- There are more chances of data inconsistency when distinct versions of the same data appear at different places.

Better Data Integration

- This system ensures better integration of data by providing a wider access to it.

Improved Data Access

- DBMS provides easy and very efficient data access to users by storing data in a well sorted manner.

Enhanced Decision Making

- It has enhanced the decision making of organization by offering a wider access to a better managed set of data.
- User are able to produce a good-quality of information when the underlying data is managed efficiently.

Improved End-User Productivity

- DBMS enables users to make quick and informed decisions in the presence of accurate data.

8. Write a note on different types of Database.

1) Centralized Database

- It is the type of database that stores data at a centralized database system.
- It comforts the users to access the stored data from different locations through several applications.

2) Distributed Database

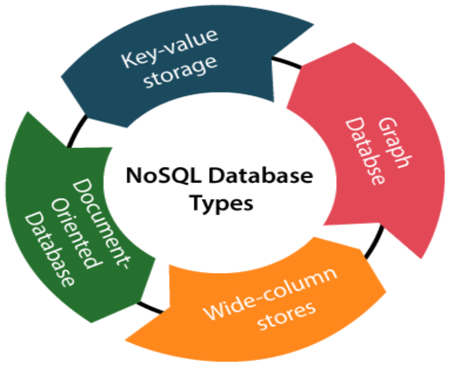
- Distributed systems, data is distributed among different database systems of an organization. **Types** : Homogeneous DDB & Heterogeneous DDB

3) Relational Database

- A relational database uses SQL for storing, manipulating, as well as maintaining the data. Each table in the database carries a key that makes the data unique from others.

4) NoSQL Database

- Non-SQL/Not Only SQL is a type of database that is used for storing a wide range of data sets.
- It is not a relational database as it stores data not only in tabular form but in several different ways. ● four types:



5) Cloud Database

- A type of database where data is stored in a virtual environment and executes over the cloud computing platform.
- It provides users with various cloud computing services (SaaS, PaaS, IaaS, etc.) for accessing the database.

6) Object-oriented Databases

- The type of database that uses the object-based data model approach for storing data in the database system.
- The data is represented and stored as objects which are similar to the objects used in the object-oriented programming language.

7) Hierarchical Databases

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- It is the type of database that stores data in the form of parent-children relationship nodes.
- it organizes data in a tree-like structure. Data get stored in the form of records that are connected via links.
- Each child record in the tree will contain only one parent. On the other hand, each parent record can have multiple child records.

8) Network Databases

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It is the database that typically follows the network data model.

- The representation of data is in the form of nodes connected via links between them.

9) Personal Database

- Collecting and storing data on the user's system defines a Personal Database. This database is basically designed for a single user.

10) Operational Database

- The type of database which creates and updates the database in real-time.
- It is basically designed for executing and handling the daily data operations in several businesses.

11) Enterprise Database

- Large organizations or enterprises use this database for managing a massive amount of data.
- It helps organizations to increase and improve their efficiency. ● Such a database allows simultaneous access to users.

9. Write a note on Relational Database Management System.

- RDBMS stands for Relational Database Management System.
- RDBMS is the basis for SQL, and for all modern database systems like MS SQL Server, IBM DB2, Oracle, MySQL, and Microsoft Access.
- The data in an RDBMS is stored in database objects which are called as tables. This table is basically a collection of related data entries and it consists of numerous columns and rows.
- Every table is broken up into smaller entities called fields. A field is a column in a table that is designed to maintain specific information about every record in the table.

advantages of RDBMS:

- It is easy to use.
- It is secured in nature.
- The data manipulation can be done.

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- It limits redundancy and replication of the data.
  It offers better data integrity.
- It provides better physical data independence.
- It provides better backup and recovery procedures.
- It provides multiple interfaces.
- Multiple users can access the database which is not possible in DBMS.

disadvantages of RDBMS:

- Software is expensive.
- Complex software refers to expensive hardware and hence increases overall cost to avail the RDBMS service.
- It requires skilled human resources to implement.
- Certain applications are slow in processing.
- It is difficult to recover the lost data.

10.Write a note on Database Language.

- **Database Languages** are the set of statements, that are used to define and manipulate a database.
- **Database Languages** are the set of statements that are used to define and manipulate a

database. A Database language has **Data Definition Language** (DDL), which is used to construct a database & it has **Data Manipulation Language** (DML), which is used to access a database.

Data Definition Language (DDL)

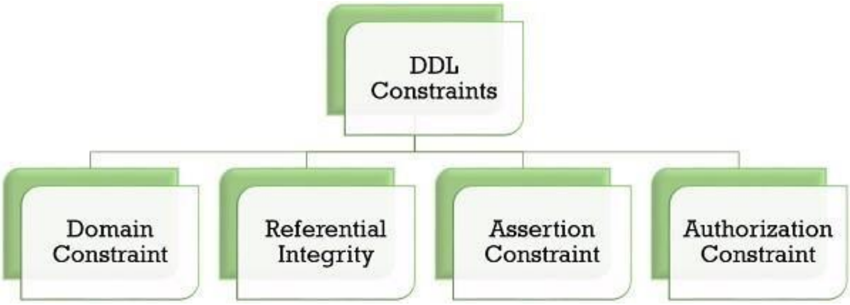
- DDL defines the statements to implement the database schema.
- DDL specifies all the schemas, i.e. physical, Logical and view schema.
- The set of statements in DDL used to implement database schema are as follow:

CREATE: This command is used to construct a relation (table) in the database.

ALTER: This command is used to reconstruct the data in the database.
DROP: This command is used to **delete** a relation in the database or an entire database.

TRUNCATE: This command **deletes all the entries** from the relation but keeps the relation structure secured in the database.

RENAME: This command **renames** the relation in a database.



Data Manipulation Language (DML)

- Data Manipulation Language has a set of statements that allows users to access and manipulate the data in the database.
- Using DML statements user can **retrieve**, **insert**, **delete** or **modify** the information in the database.
- The Data Manipulation Languages are furtherof two types, **procedural** and **non-procedural** languages:

(i) Procedural DMLs:

- Procedural DMLs are considered to be **low-level** languages, and they define **what** data is needed and **how** to obtain that data.

(ii) Non-Procedural DMLs:

- Non-Procedural DMLs are **high-level languages,** and they precisely define **what** data is required without specifying the way to access it.

some statements of DML:

SELECT: This command **reads** and pulls out the records from the database.

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INSERT: This command **adds** new records to the database.

UPDATE: This command **modifies** the data in the database.

DELETE: This command **deletes** the records in the database.

PART - C

1. Define DBMS? Explain in detail the different DBMS.

- Database Management System (DBMS) is a software for storing and retrieving users' data while considering appropriate security measures.
- It consists of a group of programs which manipulate the database.
- The DBMS accepts the request for data from an application and instructs the operating system to provide the specific data.
- In large systems, a DBMS helps users and other third-party software to store and retrieve data.
- DBMS is a collection of programs used for managing data and simultaneously it supports different types of users to create, manage, retrieve, update and store information.

Types of DBMS

The types of DBMS based on data model are as follows −

- Relational database.
- Object oriented database.
- Hierarchical database.
- Network database. **Relation**

Database

- A relational database management system (RDBMS) is a system where data is organized in two-dimensional tables using rows and columns.
- This is one of the most popular data models which is used in industries. It is based on SQL.
- Every table in a database has a key field which uniquely identifies each record.
- This type of system is the most widely used DBMS.
- Relational database management system software is available for personal computers, workstation and large mainframe systems.

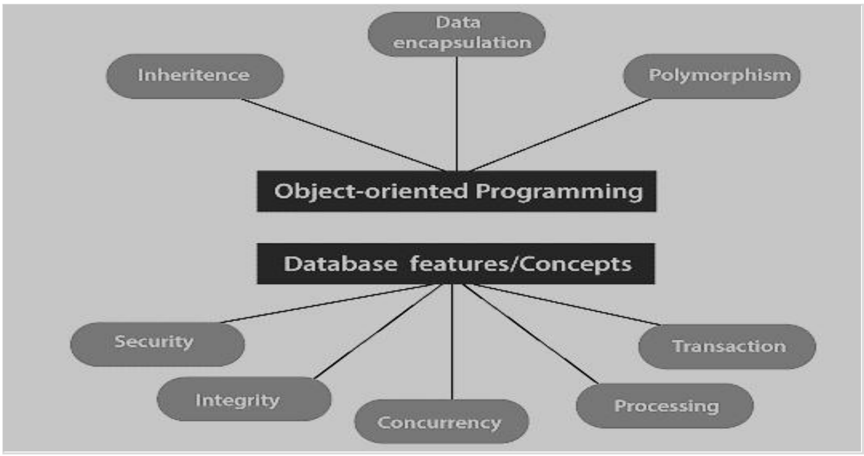
Std ID	**Name**	**City**
201	Bob	Hyderabad
204	Lucky	Chennai

205	Pinky	Bangalore

Object Oriented Database

It is a system where information or data is represented in the form of objects which is used in object-oriented programming.

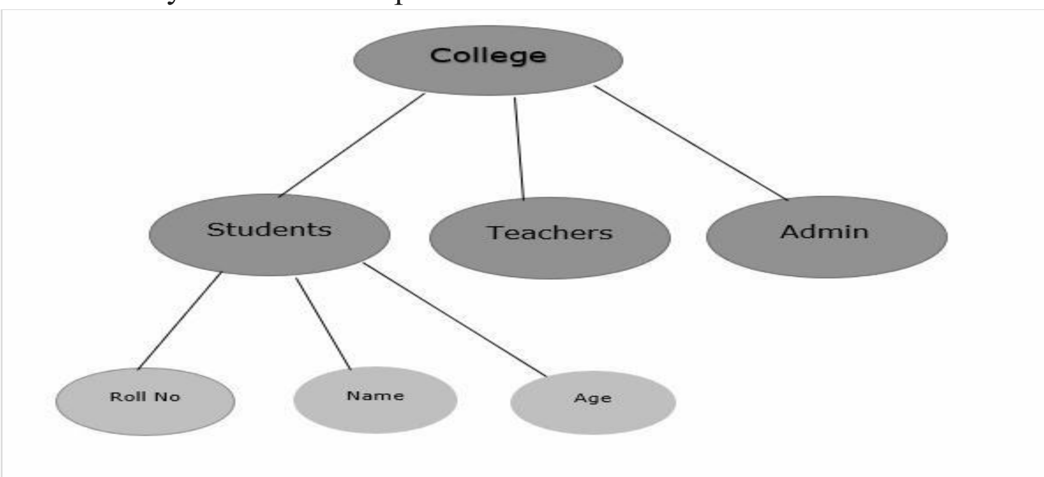
- It is a combination of relational database concepts and object-oriented principles.
- Relational database concepts are concurrency control, transactions, etc.
- OOPs principles are data encapsulation, inheritance, and polymorphism.
- It requires less code and is easy to maintain.



Hierarchical Database

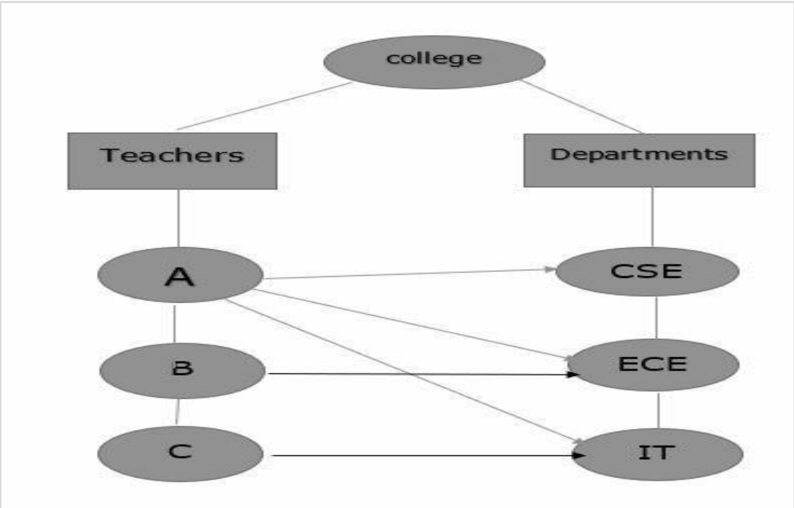
It is a system where the data elements have a one to many relationship (1: N). Here data is organized like a tree which is similar to a folder structure in your computer system.

- The hierarchy starts from the root node, connecting all the child nodes to the parent node.
- It is used in industry on mainframe platforms.



Network database

- A Network database management system is a system where the data elements maintain one to one relationship (1: 1) or many to many relationship (N: N).
- It also has a hierarchical structure, but the data is organized like a graph and it is allowed to have more than one parent for one child record.



college
Teachers
Departments
A
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CSE
ECE
IT

2. Differentiate Distributed and Centralized Database.

Difference between Centralized database and Distributed database :

Centralized database	Distributed database
It is a database that is stored, located as well as maintained at a single location only.	It is a database which consists of multiple databases which are connected with each other and are spread across different physical locations.
The data access time in the case of multiple users is more in a centralized database.	The data access time in the case of multiple users is less in a distributed database.
The management, modification, and backup of this database are easier as the entire data is present at the same location.	The management, modification, and backup of this database are very difficult as it is spread across different physical locations.
This database provides a uniform and complete view to the user.	Since it is spread across different locations thus it is difficult to provide a uniform view to the user.
This database has more data consistency in comparison to distributed database.	This database may have some data replications thus data consistency is less.
The users cannot access the database in case of database failure occurs.	In distributed database, if one database fails users have access to other databases.

Centralized database is less costly.	This database is very expensive.

3. Explain in detail the significance of Database with its applications?

Improved data sharing and data security
Database management systems help users share data quickly, effectively, and securely across an organization. By providing quick solutions to database queries, a data management system enables faster access to more accurate data. End users, like salespeople, are able to speed up sales cycles and get more accurate in their sales prospecting.

Effective data integration
Implementing a database management system will promote a more integrated picture of your operations by easily illustrating how processes in one segment of the organization affect other segments. What once was done completely manually now can be fully automated and more accurate. The right DBMS will include flexible integration options to standardize data across multiple sources, remove duplicates, normalize, segment, and enrich data sets into custom workflows.

Consistent, reliable data
Data inconsistency occurs when different versions of matching data exist in different places in an organization. For example, one group has a client's correct email, another the correct phone number. By using a proper database management system and data quality tools, you can be sure that an accurate view of data is shared throughout your organization.

Data that complies with privacy regulations
Database management systems provide a better framework for the enforcement of privacy and security policies. By orchestrating data in a unified manner, companies can manage privacy and data security centrally, helping unify their systems of record and lower the risk of regulatory violations.

Increased productivity
Deploying a DBMS typically results in increased productivity because a good DBMS empowers people to spend more time on high-value activities and strategic initiatives, and less time cleaning data and manually scrubbing lists.

Better decision-making
Decisions built on data are only as good as the information used. A database management system helps provide a framework to facilitate data quality initiatives. Better data management procedures generate higher-quality information, which leads to better decision-making across an organization.

Advantages of Database Management System (DBMS):

Some of them are given below.

Better Data Transferring: Database management creates a place where users have an advantage of more and better-managed data. Thus making it possible for end-users to have a quick look and to respond fast to any changes made in their environment.

Better Data Security: The more accessible and usable the database, the more it is prone to security issues. As the number of users increases, the data transferring or data sharing rate also increases thus increasing the risk of data security.

Better data integration: Due to the Database Management System we have an access to well managed and synchronized form of data

Minimized Data Inconsistency: Data inconsistency occurs between files when different versions of the same data appear in different places.

Faster data Access: The Database management system (DBMS) helps to produce quick answers to database queries thus making data access faster and more accurate. For example, to read or update the data.

Better decision making: Due to DBMS now we have Better managed data and Improved data access because of which we can generate better quality information hence on this basis better decisions can be made.

Increased end-user productivity: The data which is available with the help of a combination of tools that transform data into useful information, helps end-users to make quick, informative, and better decisions that can make difference between success and failure in the global economy.

Simple: Database management system (DBMS) gives a simple and clear logical view of data. Many operations like insertion, deletion, or creation of files or data are easy to implement.

Data abstraction: The major purpose of a database system is to provide users with anabstract view of the data.

Reduction in data Redundancy: When working with a structured database, DBMS provides the feature to prevent the input of duplicate items in the database.

4.**Explain in detail the Distributed database.**

- A distributed database is a collection of multiple interconnected databases, which are spread physically across various locations that communicate via a computer network.

Features

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Advantages

- **Modular Development** − If the system needs to be expanded to new locations or new units, in centralized database systems, the action requires substantial efforts and disruption in the existing functioning.
- **More Reliable** − In case of database failures, the total system of centralized databases comes to a halt.
- **Better Response** − If data is distributed in an efficient manner, then user requests can be met from local data itself, thus providing faster response.
- **Lower Communication Cost** − In distributed database systems, if data is located locally where it is mostly used, then the communication costs for data manipulation can be minimized. This is not feasible in centralized systems.

Disadvantages

- Costly Software. Ensuring data transparency and coordination across multiple sites often requires using expensive software in a distributed database system.
- Large Overhead. Many operations on multiple sites requires numerous calculations and constant synchronization when database replication is used, causing a lot of processing overhead.
- Data Integrity. A possible issue when using database replication is data integrity, which is compromised by updating data at multiple sites.
- Improper Data Distribution. Responsiveness to user requests largely depends on proper data distribution. That means responsiveness can be reduced if data is not correctly distributed across multiple sites.

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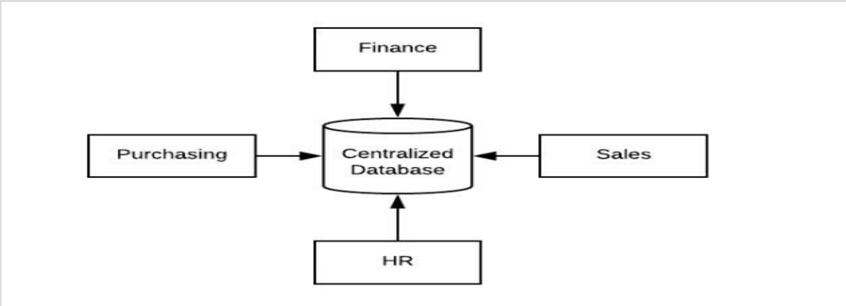
4. **Explain in detail Centralized Database.**

- A centralized database is stored at a single location such as a mainframe computer. It is maintained and modified from that location only and usually accessed using an internet connection such as a LAN or WAN.
- The centralized database is used by organizations such as colleges, companies, banks etc.



Functions of centralized database

Distributed query processing

The basic function of centralized database management system is to provide facility and give access to all the connected computers which fulfill all requirements requested by any single node.

Single central unit

All the data and information are stored in single centralized database management system. The computer system which fulfills the requirements of all the connected computers is known as server and other computers are known as clients.

Transparency

All the queries are processed in a single computer system also known as server. There is no duplication or irrelevant data stored in this management system. All connected computer has the access to central computer for their query processing and requirement.

Scalable

No of computers can be added in this centralized database management system. These computers are connected to the system through a network.

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Advantages

- The data integrity is maximised as the whole database is stored at a single physical location. This means that it is easier to coordinate the data and it is as accurate and consistent as possible.
  The data redundancy is minimal in the centralized database.
- Since all the data is in one place, there can be stronger security measures around it. So, the centralized database is much more secure.
- Data is easily portable because it is stored at the same place.

Disadvantages

- Since all the data is at one location, it takes more time to search and access it. If the network is slow, this process takes even more time.
- There is a lot of data access traffic for the centralized database. This may create a bottleneck situation.
- Since all the data is at the same location, if multiple users try to access it simultaneously it creates a problem. This may reduce the efficiency of the system.
- If there are no database recovery measures in place and a system failure occurs, then all the data in the database will be destroyed.